CODE APEX

THE END IS COMING.

verzii

Team Details

- Team name - RavenClaw
- Team leader Name - Devraj Sundrani
- Problem Statement (Domain) - Open Innovation

Problem Statement & Target Need

Food & drug compliance remains a high-stakes trust problem because verification is manual, fragmented and largely inaccessible to the people most affected.

01 Who is affected ?



Consumers

Cannot verify what they buy



Vulnerable Groups

Higher risk from hidden or undeclared ingredients



Enforcement Officers

Overworked manual checks with inconsistent results



Businesses & Regulators

Honest players undercut; limited visibility & violations

02 Why it matters now ?

20 MIN

Manual Check

~ **2 MIN**

GUIDED SCAN

REACTIVE

One complaint at a time

RISK-BASED

PRIORITIZED ENFORCEMENT

ISOLATED

Handwritten Findings

TRACEABLE

EVIDENCE-BACKED CASES

PASSIVE

Consumer as Victim

PARTICIPATORY

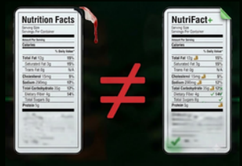
CONSUMER AS ENFORCEMENT PARTNER

03 The current gap



COVERAGE GAP

Too many products, too few inspections



CONSISTENCY GAP

Same label, different interpretation



CITIZEN GAP

No easy way to detect, prove or report



COMPREHENSION GAP

Dense and misleading labels are hard to understand

04 Target need



Scan

Capture complete package



Verify

Extract declarations and apply legal rules



Prove

Generate rule-cited, tamper-evident evidence

Act

Route to authorities

Proposed Solution & Innovation

NIRIKSHAK AI- An AI-guided platform that turns a simple product scan into a **trustworthy** rule-cited compliance verdict and consumer health insight.

Core Features

01 Guided Capture

Scan all sides of the package with on-screen guidance.



- Front
- Back
- Side
- Top

02 AI Extraction

OCR + NLP extracts key declarations.



- Ingredients
- Nutrition Table
- Expiry Date
- MRP (Sticker/Printed)
- Manufacturer / FSSAI
- Claims & Symbols

03 Rule-Based Verification

Checks against Food Safety, Legal Metrology and Drug rules.

- Banned ingredients
- Allergen detection
- Nutrition verification
- FSSAI licence check
- MRP compliance
- Misleading claims / ads

04 Insights & Benchmarking

Flags dark patterns, surrogate ads and compares nutrition with healthy benchmarks.

Potential Issues

- Uses 'Immunity Boost' (claim without basis)
- High sugar vs. healthy benchmark
- Small MRP sticker over original MRP

Health Context

Sugar
Sodium
Fat
Serving size

05 Verdict & Report

Plain-language summary, 5-star rating and detailed rule-cited report.



Overall Rating

High Risk

Key Issues

- Banned additive detected
- Excess sodium
- MRP sticker over original MRP
- Misleading claim

USP & Innovation

Banned & Allergen Detection

Checks ingredients for banned additives and allergy-causing components.

Dark Pattern & Surrogate Ad Detection

Identifies misleading symbols, false claims and surrogate ads

Nutrition Benchmarking

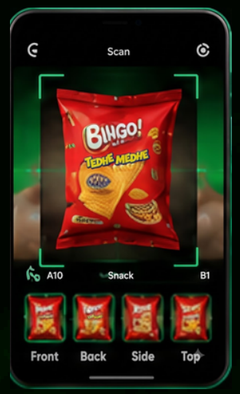
Compares nutrition table with healthy benchmarks and flags high-risk products.

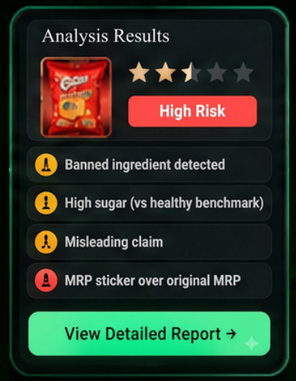
Actionable Reports

Clear, rule-cited reports with reasons, evidence and a 5-star consumer rating.

SOLUTION EXPERIENCE

From scan to clear insights in seconds







Feasibility

Leverages mature, open-source tools

PaddleOCR, LLMs, vector search and proven rule engines available.

Focused MVP scope

Core scan → extract → verify → report flow with key rules (FSSAI, Legal Metrology).

Accessible data & rule sets

Publicly available regulations, sample products and benchmarks for prototyping.

System Architecture

1. Multi-Channel Capture Layer (Mobile / Web – Guided Capture)

On-Device Validation
Complete coverage check
Guidance prompts



- Front
- Back
- Side
- Top/ Bottom

1. Captures Image

2. AI/ML & Extraction Layer (From Image to Structured Data)

Image Pre-processing
Enhance, dewarp, denoise

OCR (PaddleOCR)
Extract text from all sides

NLP & Field Extraction
Ingredients, nutrition table, expiry, MRP, manufacturer, FSSAI etc.

Auxiliary Models (USP)
Allergen/banned ingredient detection
Dark pattern & surrogate ad detection
Symbol and claim interpretation
Nutrition table parsing

3. Legal Core + Rule Engine Layer (Decides Law)

FSSAI Rules | Legal Metrology Rules | Drug Rules

Rule Engine
Versioned, deterministic rules
Checks each declaration
Flags violations and risky claims
Handles rule updates

LLM / RAG (Optional for reasoning)
Retrieve relevant rule clauses
Explain reasoning in plain language
Natural-language queries for officers

4. Evidence & Human Review Layer (Expert Validation)

Risk & Health Analysis
Compliance & risk score
Compare nutrition with healthy benchmarks
Identify high-risk ingredients
Surrogate ad & dark pattern analysis

Verdict Generation
Rule-cited explanation
5-star (Likert) consumer rating
Highlight key issues
Compare with healthy benchmarks

Data Repository & Analytics
Store scans, results, images, rules
Searchable database for officers
Macro-level trends, hotspots, repeat offenders

5. Search & Citizen Services Layer (Access & Impact)

- PostgresSQL ↕ MongoDB
- Incentive Ledger
- Grievance Redressal
- Trust Portal ↕ DigitalTwin ↕ Geographic Dashboard

Users

- Consumer
- JWT (Role Based Access)
- Inspector (Field)

Tech Stack

- **Redis** — Cache & Quick
- **FastAPI** — Backend
- **LangChain** — Orchestration
- **ChromaDB** — VectorDB
- **PaddleOCR** — Text Extraction
- **Llama 3** — LLM verifier
- **Flutter** — Mobile App
- **Next.js** — Website

Key Features & UI/UX

01 AI-Powered Smart Scanner

The Eyes

Scans food and medicine packages using OCR/NLP to extract ingredients, expiry, MRP, nutrition and licence details.

Key feature:
Multi-angle capture and potential tampering detection.

02 AI Compliance Engine

The Brain

Checks labels against updated legal rules for ingredient, allergen, expiry and labelling issues.

Key feature:
Flags misleading terminology and dark patterns for human review.

03 Consumer Safety & Trust Portal

The Voice

Provides clear safety reports, allergy/nutrition alerts, product ratings and comparisons.

Key feature:
Flags physical-vs-online price and quantity mismatches.

04 Citizen Grievance & Intelligent Enforcement

The Muscle

Turns citizen complaints into trackable cases with evidence and officer verification.

Key feature:
Product history, risk prioritization and geographic violation insights.



Implementation Roadmap & Scope

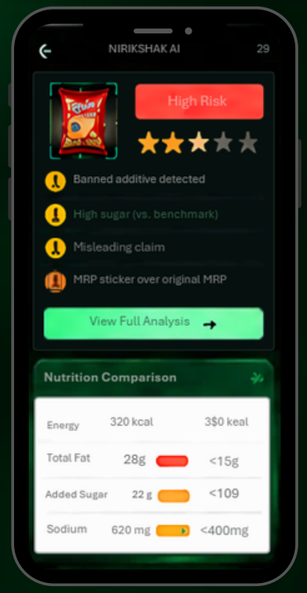
From a working hackathon prototype to a scalable, real-world compliance platform.

Phase 1 — Built Today



- Multi-angle product capture (front, back, sides)
- OCR extraction of key fields (ingredients, nutrition, expiry, MRP, FSSAI, manufacturer)
- Rule-based compliance check (food safety, legal metrology)
- Detect banned/allergen ingredients and dark patterns (initial version)
- 5-star consumer rating with plain-language explanation
- Detailed report generation (rule-cited with evidence)

Phase 2 — In Progress



- Improved OCR & NLP (more languages, more formats)
- Nutrition benchmarking against healthy standards
- Surrogate ad detection and advanced dark pattern analysis
- Natural-language queries for officers(e.g., "show all non-compliant biscuit brands in Pune this month")
- Officer dashboard (inspections, case management, analytics)
- Enhanced model accuracy and new product categories (medicines, cosmetics)

Phase 3 — Planned Beyond Hackathon



- Mobile app for consumers (scan, get verdict, reports)
- Full regulatory integration (e-filing, eProsecution)
- Macro-level analytics & risk scoring (heatmaps, repeat offenders, trends)
- Wide product coverage (food, drugs, cosmetics, other packaged goods)
- Incentive & credibility system (for verified citizen reports)
- Nationwide rollout with government partnerships

Scope

In Scope

- Packaged food and drug products
- Label-based compliance verification
- Consumer and officer-facing platform
- Evidence-backed reports and analytics

Out of Scope (Hackathon)

- Laboratory testing for chemical contaminants
- Unpackaged /loose products
- Complete regulatory integration
- Nationwide rollout and large-scale deployment

Development Approach

Modular development → Agile iterations → Real data & test cases → Human-in-the-loop validation

Expected Impact & Scalability

From everyday scans to system-wide trust



Consumers

- Plain-language health verdict
- Ingredient & allergen warnings
- 5-star trust rating
- Easy reporting with rewards



Enforcement Officers

- Better allocation of resources
- Track repeat offenders
- Rule-cited, tamper-evident reports
- Guided 2-minute checks (vs ~20 minutes manual)



Businesses & Retailers

- Fewer false accusations
- Fair competition in the market
- Stand out with compliance badge
- Build trust through compliance



Policy Makers & Regulators

- Safer food and drug supply chains
- Data-driven policy and enforcement
- Trend & hotspot analysis
- Real-time view of non-compliance

Key Impact Areas

Public Health

Reduces risk from adulterated, mislabeled and unsafe products

Economic Impact

Less wastage, fair pricing and stronger market for compliant businesses

Enforcement Efficiency

Faster, standardized and auditable inspections

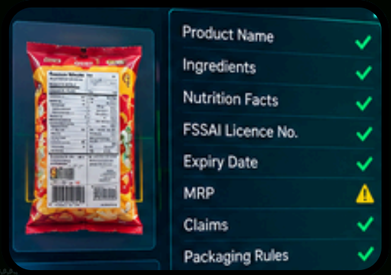
Citizen Engagement

Turns consumers into active partners in safety

From a label scan to real-world impact



1 Scan Product
Capture all sides of the package



2 Extract & Verify
AI reads label and checks against rules



3 Flag & Review
Clear explanation with rule reference



4 Human Decision
Officer verifies and takes action



5 Lab Testing (if needed)
High-risk products sent for sampling



6 Action & Feedback
Penalties, recalls or advisory. Feeds back to improve system.

VISHWAKARMA INSTITUTES VI

AISF
ARTIFICIAL INTELLIGENCE STUDENT FORUM

VCA

CODE APEX

BUILD BEFORE IT DOES.

POWERED BY verzii

THANK YOU